

Temă 2 - M5

Ecuații diferențiale de ordin superior cu coeficienți constanți:

$$1) y''' + y = 0.$$

Ec. caracteristică: $\pi^3 + 1 = 0$

Schema lui Horner:

$$\begin{array}{r|rrrrr} & 1 & 0 & 0 & 1 & \\ -1 & 1 & -1 & 1 & 0 & \end{array} \rightarrow \pi_1 = -1 \text{ rădăcină reală distinctă}$$

$$\pi^2 - \pi + 1 = 0$$

$$\Delta = b^2 - 4ac = 1 - 4 = -3$$

$$\Delta < 0 \rightarrow \pi_{2,3} = \frac{-b \pm i\sqrt{-\Delta}}{2a} \rightarrow \begin{cases} \pi_2 = \frac{1}{2} + \frac{\sqrt{3}}{2}i \\ \pi_3 = \frac{1}{2} - \frac{\sqrt{3}}{2}i \end{cases} \left. \begin{array}{l} \text{rădăcini} \\ \text{complexe} \\ \text{conjugate} \end{array} \right\}$$

$$\alpha = \operatorname{Re}(\pi_2) = \frac{1}{2}$$

$$\beta = \operatorname{Im}(\pi_2) = \frac{\sqrt{3}}{2}$$

$$y_1(x) = e^{\pi_1 x} = e^{-x}$$

$$y_2(x) = e^{\alpha x} \cos \beta x = e^{\frac{x}{2}} \cos\left(\frac{\sqrt{3}}{2}x\right)$$

$$y_3(x) = e^{\alpha x} \sin \beta x = e^{\frac{x}{2}} \sin\left(\frac{\sqrt{3}}{2}x\right)$$

Se arată că $\forall (y_1, y_2, y_3) \neq 0 \Rightarrow y_1, y_2, y_3$ - liniar independente $\rightarrow y_1, y_2, y_3$ - sistem fundamental de soluții.

$$y_0(x) = c_1 y_1(x) + c_2 y_2(x) + c_3 y_3(x)$$

$$y_0(x) = c_1 \cdot e^{-x} + c_2 \cdot e^{\frac{x}{2}} \cos\left(\frac{\sqrt{3}}{2}x\right) + c_3 \cdot e^{\frac{x}{2}} \sin\left(\frac{\sqrt{3}}{2}x\right),$$

$$c_1, c_2, c_3 \in \mathbb{R}.$$

$$2) y''' - 3y'' + 3y' - y = 0$$

$$\text{Ec caracteristică: } \pi^3 - 3\pi^2 + 3\pi - 1 = 0$$

$$\begin{array}{r|rrrr} & \pi^3 & -3\pi^2 & +3\pi & -1 \\ 1 & 1 & -2 & 1 & 0 \end{array} \Rightarrow \pi_1 = 1$$

$$\pi^2 - 2\pi + 1 = 0$$

$$\Delta = b^2 - 4ac = 4 - 4 = 0 \Rightarrow \pi_2 = \pi_3 = \frac{2}{2} = 1$$

adaună
rădăna triplă

$$y_1(x) = e^{\pi_1 x} = e^x$$

$$y_2(x) = x e^{\pi_2 x} = x e^x$$

$$y_3(x) = x^2 e^{\pi_3 x} = x^2 e^x$$

Se arată că $\omega(y_1, y_2, y_3) \neq 0 \Rightarrow y_1, y_2, y_3$ linear independente $\Rightarrow y_1, y_2, y_3$ - sistem fundamental de soluții.

$$y_0(x) = c_1 y_1(x) + c_2 y_2(x) + c_3 y_3(x)$$

$$y_0(x) = c_1 \cdot e^x + c_2 x e^x + c_3 x^2 e^x,$$

$$c_1, c_2, c_3 \in \mathbb{R}.$$

$$3) y^{(4)} - 5y'' + 4y = 0$$

$$\text{Ecuația caracteristică: } \pi^4 - 5\pi^2 + 4 = 0.$$

Schema lui Horner:

$$\begin{array}{r|rrrrr} & 1 & 0 & -5 & 0 & 4 \\ 1 & 1 & 1 & -4 & -4 & 0 \end{array} \rightarrow \pi_1 = 1$$

$$\pi^3 + \pi^2 - 4\pi - 4 = 0$$

$$\begin{array}{r|rrrr} & 1 & 1 & -4 & -4 \\ -1 & 1 & 0 & -4 & 0 \end{array} \rightarrow \pi_2 = -1$$

$$\pi^2 - 4 = 0 \rightarrow \pi_3 = 2, \pi_4 = -2$$

radăcini
reale
distinse

$$y_1(x) = e^{\pi_1 x} = e^x$$

$$y_2(x) = e^{\pi_2 x} = e^{-x}$$

$$y_3(x) = e^{\pi_3 x} = e^{2x}$$

$$y_4(x) = e^{\pi_4 x} = e^{-2x}$$

Se arată că $w(y_1, y_2, y_3, y_4) \neq 0 \rightarrow$

y_1, y_2, y_3, y_4 - liniar independente $\rightarrow$

y_1, y_2, y_3, y_4 - sistem fundamental de soluții.

$$y_0(x) = c_1 y_1(x) + c_2 y_2(x) + c_3 y_3(x) + c_4 y_4(x)$$

$$y_0(x) = c_1 e^x + c_2 e^{-x} + c_3 e^{2x} + c_4 e^{-2x},$$

$$c_1, c_2, c_3, c_4 \in \mathbb{R}.$$

$$4. \quad y''' - 6y'' + 12y' + y = 0$$

$$\text{Ec. caract: } \pi^3 - 6\pi^2 + 12\pi + 1 = 0$$

CERINȚĂ GREȘITĂ

$$5) y''' - y = \sin x + 3x \cos x.$$

Ec. caracteristică: $\pi^3 - 1 = 0$.

Schema lui Horner:

$$\begin{array}{r|rrrr} & 1 & 0 & 0 & -1 \\ 1 & 1 & 1 & 1 & 0 \end{array} \rightarrow \pi_1 = 1 \text{ - rădăcină reală distinctă}$$

$$\pi^2 + \pi + 1 = 0.$$

$$\Delta = b^2 - 4ac = 1 - 4 = -3$$

$$\pi_{2,3} = \frac{-b \pm i\sqrt{-\Delta}}{2a} = \frac{-1 \pm i\sqrt{3}}{2} \rightarrow \pi_2 = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$$

$$\rightarrow \pi_3 = -\frac{1}{2} - \frac{\sqrt{3}}{2}i$$

rădăcini
complexe
conjugate

$$\alpha = \operatorname{Re}(\pi_2) = -\frac{1}{2}$$

$$\beta = \operatorname{Im}(\pi_2) = \frac{\sqrt{3}}{2}$$

$$y_1(x) = e^{\pi_1 x} = e^x$$

$$y_2(x) = e^{\alpha x} \cos \beta x = e^{-\frac{x}{2}} \cos\left(\frac{\sqrt{3}}{2}x\right)$$

$$y_3(x) = e^{\alpha x} \sin \beta x = e^{-\frac{x}{2}} \sin\left(\frac{\sqrt{3}}{2}x\right)$$

Se arată că $w(y_1, y_2, y_3) \neq 0 \rightarrow y_1, y_2, y_3$ - liniar independente $\Rightarrow y_1, y_2, y_3$ sistem fundamental de soluții.

$$y_0(x) = C_1 \cdot y_1(x) + C_2 \cdot y_2(x) + C_3 \cdot y_3(x)$$

$$y_0(x) = C_1 \cdot e^x + C_2 \cdot e^{-\frac{x}{2}} \cos\left(\frac{\sqrt{3}}{2}x\right) + C_3 \cdot e^{-\frac{x}{2}} \sin\left(\frac{\sqrt{3}}{2}x\right)$$

$$f(x) = e^{\alpha x} [P(x) \cos \beta x + Q(x) \sin \beta x]$$

$$f(x) = \sin x + 3x \cos x.$$

$$\hookrightarrow \alpha = 0, \beta = 1, P(x) = 3x, Q(x) = 1.$$

$\alpha + i\beta = 0 + i = i$ - nu este rădăcină a ec. caract.

$$y_p = e^{ax} [A(x) \cos bx + B(x) \sin bx]$$

$$g_1(A, B) = \max(g_1(P, Q)) = \max(1, 0) = 1$$

$$g_p(x) = (ax+b) \cos x + (cx+d) \sin x$$

$$y_p'(x) = (-ax+c-b) \sin x + (cx+d+a) \cos x$$

$$y_p''(x) = -(cx+d+2a) \sin(x) - (ax-2c+b) \cos x$$

$$y_p'''(x) = (ax-3c+b) \sin x + (-cx-d-3a) \cos x$$

$$(ax-3c+b) \sin x + (-cx-d-3a) \cos x - (ax+b) \cos x - (cx+d) \sin x = \sin x + 3x \cos x$$

$$\sin x: ax-3c+b-cx-d = 1$$

$$\cos x: -cx-d-3a-ax-b = 3x$$

$$y_g = y_0 + y_p = c_1 e^x + c_2 e^{-\frac{x}{2}} \cos \frac{\sqrt{3}}{2} x + c_3 e^{-\frac{x}{2}} \sin \frac{\sqrt{3}}{2} x + (ax-3c+b-cx-d) \sin x + (-cx-d-3a-ax-b) \cos x$$

$$6) y''' - 3y' + 2y = x^2 + 3x + 4.$$

$$r^3 - 3r + 2 = 0$$

$$Ec \text{ omogenă: } y''' - 3y' + 2y = 0.$$

$$Ec \text{ caracter: } r^3 - 3r + 2 = 0$$

Horner:

$$\begin{array}{r|rrrr} & 1 & 0 & -3 & 2 \\ 1 & 1 & 1 & -2 & 0 \end{array} \rightarrow r_1 = 1.$$

$$r^2 + r - 2 = 0.$$

$$\Delta = b^2 - 4ac = 1 + 8 = 9 \rightarrow r_2 = \frac{-1+3}{2} = \frac{2}{2} = 1$$

$$\rightarrow r_3 = \frac{-1-3}{2} = \frac{-4}{2} = -2$$

$r_1 = r_2 = 1$ - rădăcină reală dublă

$r_3 = -2$ - rădăcină reală distinctă

$$\begin{aligned} y_1(x) &= e^{r_1 x} = e^x \\ y_2(x) &= x e^{r_2 x} = x e^x \\ y_3(x) &= e^{-2x} \end{aligned} \left. \begin{array}{l} \\ \\ \end{array} \right\} \begin{array}{l} \rightarrow \text{Se arată că } W(y_1, y_2, y_3) \neq 0 \Rightarrow \\ y_1, y_2, y_3 \text{ - liniar independente} \Rightarrow \\ y_1, y_2, y_3 \text{ - sistem fundamental de soluții.} \end{array}$$

$$f(x) = x^2 + 3x + 4 = e^{xx} \cdot P_m(x)$$

$x=0$ - nu este rădăcină

$$y_{PN} = e^{xx} \cdot Q_m(x), \quad Q_m - \text{polinom arbitrar.}$$

$$P(x) = x^2 + 3x + 4 \rightarrow Q_m(x) = ax^2 + bx + c. \text{ verifică ec.}$$

$$(ax^2 + bx + c)^{(3)} - 3(ax^2 + bx + c)' + 2 \cdot (ax^2 + bx + c) = x^2 + 3x + 4.$$

$$2ax^2 + 2bx - 6ax + 2c - 3b = x^2 + 3x + 4.$$

$$x^2: 2a = 1 \rightarrow a = \frac{1}{2}$$

$$x^1: 2b - 6a = 3 \rightarrow 2b - 3 = 3 \rightarrow 2b = 6 \rightarrow b = 3$$

$$x^0: 2c - 3b = 4 \rightarrow 2c - 9 = 4 \rightarrow 2c = 13 \rightarrow c = \frac{13}{2}$$

$$y_G = \underbrace{C_1 \cdot e^x + C_2 \cdot x e^x + C_3 \cdot e^{-2x}}_{y_0} + \underbrace{\frac{1}{2} x^2 + 3x + \frac{13}{2}}_{y_{PN}}, \quad C_1, C_2, C_3 \in \mathbb{R}.$$

$$7) y''' + 8y = e^{-2x}$$

$$\text{Ec. omogenă: } r^3 + 8 = 0$$

$$\text{Ec. caracteristică: } r^3 = -8$$

Harmur:

$$\begin{array}{c|cccc} & 1 & 0 & 0 & 8 \\ \hline -2 & 1 & -2 & 4 & 0 \end{array} \rightarrow r_1 = -2 \text{ rădăcină reală distinctă}$$

$$r^2 - 2r + 4 = 0$$

$$\Delta = b^2 - 4ac = 4 - 16 = -12$$

$$r_{2,3} = \frac{-b \pm i\sqrt{-\Delta}}{2a} = \frac{2 \pm i\sqrt{12}}{2} \rightarrow \begin{cases} r_2 = 1 + \sqrt{3}i \\ r_3 = 1 - \sqrt{3}i \end{cases} \left. \begin{array}{l} \text{rădăcini} \\ \text{complexe} \\ \text{conjugate} \end{array} \right\}$$

$$\alpha = \operatorname{Re}(r_2) = 1$$

$$\beta = \operatorname{Im}(r_2) = \sqrt{3}$$

$$y_1(x) = e^{r_1 x} = e^{-2x}$$

$$y_2(x) = e^{r_2 x} \cos \beta x = e^x \cos(\sqrt{3}x)$$

$$y_3(x) = e^{r_3 x} \sin \beta x = e^x \sin(\sqrt{3}x)$$

$\rightarrow$ Se arată că $w \neq 0$

y_1, y_2, y_3 - baze în $\mathbb{R}^3$

y_1, y_2, y_3 - sist. fund. de sol.

$$y_0(x) = c_1 \cdot e^{-2x} + c_2 \cdot e^x \cos(\sqrt{3}x) + c_3 \cdot e^x \sin(\sqrt{3}x),$$

$$c_1, c_2, c_3 \in \mathbb{R}.$$

$$f(x) = e^{-2x} = e^{xx} \cdot P_m(x)$$

$\alpha = -2$ este rădăcină a ec. caracteristice de ordin $k=3$.

$$P_m(x) = 1$$

$$y_{PN} = e^{xx} \cdot x^k \cdot P_m(x)$$

$$y_{PN} = e^{-2x} \cdot x^3$$

$$y_G = y_0 + y_{PN} = c_1 \cdot e^{-2x} + c_2 e^x \cos(\sqrt{3}x) + c_3 e^x \sin(\sqrt{3}x) + e^{-2x} \cdot x^3$$

$$c_1, c_2, c_3 \in \mathbb{R}.$$

$$8) y'' + y = \lg x$$

$$E_c \cdot \text{omogena: } y' + y = 0$$

$$E_c \cdot \text{caracteristica: } \pi^2 + 1 = 0$$

$$\pi^2 = -1 \rightarrow \pi_1 = i \quad \left. \begin{array}{l} \pi_1 = i \\ \pi_2 = -i \end{array} \right\} \begin{array}{l} \text{rădăcini complexe} \\ \text{conjugate} \end{array}$$

$$\alpha = \operatorname{Re}\{i\} = 0$$

$$\beta = \operatorname{Im}\{i\} = 1$$

$$\begin{aligned} y_1(x) &= e^{\alpha x} \cos(\beta x) = \cos x \\ y_2(x) &= e^{\alpha x} \sin(\beta x) = \sin x \end{aligned} \quad \left. \begin{array}{l} \text{Se arată că } y_1, y_2 \neq 0 \\ y_1, y_2 \text{ - funcții independente} \end{array} \right\} \rightarrow y_1, y_2 \text{ - sistem fund. de sol}$$

$$y_c(x) = C_1 \cos x + C_2 \sin x, \quad C_1, C_2 \text{ - constante.}$$

$$y_p(x) = C_1(x) \cos x + C_2(x) \sin x$$

$$\begin{cases} C_1' \cos x + C_2' \sin x = 0 \rightarrow C_1' = -\frac{C_2' \sin x}{\cos x} \\ -C_1' \sin x + C_2' \cos x = \lg x \end{cases}$$

$$+ \frac{C_2' \sin^2 x}{\cos x} + C_2' \cos x = \lg x \Rightarrow C_2' \left(\frac{\sin^2 x}{\cos x} + \cos x \right) = \lg x$$

$$C_2' \cdot \frac{\sin^2 x + \cos^2 x}{\cos x} = \lg x \rightarrow C_2' \cdot \frac{1}{\cos x} = \lg x \rightarrow C_2' = \lg x \cdot \cos x$$

$$\text{Dar } \lg x = \frac{\sin x}{\cos x} \Rightarrow C_2' = \frac{\sin x}{\cos x} \cdot \cos x \Rightarrow C_2' = \sin x \Rightarrow C_2 = \int \sin x \, dx$$

$$= -\cos x + k_2$$

$$C_1' = -\frac{C_2' \sin x}{\cos x} = -\frac{\cancel{\cos x} \cdot \sin x}{\cancel{\cos x}} = -\sin x \rightarrow C_1' = -\sin x \Rightarrow C_1 = \int -\sin x \, dx$$

$$C_1 = \cos x + k_1$$

$$y_G = y(x) = (k_1 - \cos x) \cdot \cos x + (k_2 - \cos x) \sin x, \quad k_1, k_2 \text{ - constante.}$$

$$g) y^{(4)} - 2y^{(3)} + y'' = x + e^x$$

$$\text{Ec. omogenă: } y^{(4)} - 2y^{(3)} + y'' = 0$$

$$\text{Ec. caracteristică: } \pi^4 - 2\pi^3 + \pi^2 = 0$$

$$\pi^2(\pi^2 - 2\pi + 1) = 0$$

$$\left. \begin{array}{l} \pi_1 = \pi_2 = 0 \\ \pi_3 = \pi_4 = 1 \end{array} \right\} \begin{array}{l} \text{rădăcini reale} \\ \text{duble.} \end{array}$$

$$y_1(x) = e^{\pi_1 x} = e^0 = 1$$

$$y_2(x) = x e^{\pi_2 x} = x e^0 = x$$

$$y_3(x) = e^{\pi_3 x} = e^x$$

$$y_4(x) = x e^{\pi_4 x} = x e^x$$

Se arată că $w(y_1, y_2, y_3, y_4) \neq 0 \Rightarrow$

y_1, y_2, y_3, y_4 - funcții independente

$\Rightarrow y_1, y_2, y_3, y_4$ - Sistem fundamental de soluții.

$$y_0(x) = c_1 + c_2 x + c_3 e^x + c_4 x e^x, \quad c_1, c_2, c_3, c_4 \in \mathbb{R}.$$

$$f_1(x) = x \Rightarrow \alpha = 0, \beta = 0, P(x) = x, Q(x) = 0$$

$$x + i\beta = 0 \text{ rădăcină de ordin } k=2$$

$$y_{p1}(x) = x^2(ax + b)$$

$$f_2(x) = e^x \Rightarrow \alpha = 1, \beta = 0, P(x) = 1, Q(x) = 0$$

$$x + i\beta = 1 \text{ răd de ord } k=2.$$

$$y_p(x) = x^2 \cdot e^x \cdot c$$

$$y_p(x) = y_{p1}(x) + y_{p2}(x) = ax^3 + bx^2 + cx^2 e^x$$

$$y_p' = 3ax^2 + 2bx + 2cx e^x + cx^2 e^x$$

$$y_p'' = 6ax + 2b + 2ce^x + 2cx e^x + 2cx e^x + cx^2 e^x$$

$$y_p''' = 6a + 6ce^x + 6cx e^x + cx^2 e^x$$

$$y_p^{(4)} = 12ce^x + 8cx e^x + cx^2 e^x.$$

$$\cancel{12ce^x} + \cancel{8ce^x} + \cancel{cx^2e^x} - 12a - \cancel{12ce^x} - \cancel{12cx e^x} - \cancel{2cx^2e^x} + 6ax + 2b + 2ce^x + 4cx e^x + \cancel{cx^2e^x} = x + e^x$$

$$10ce^x - 8cx e^x - 12a + 6ax + 2b = x + e^x$$

$$e^x (10c - 8cx) + 6ax - 12a + 2b = x + e^x$$

$$a \cdot x = x \rightarrow a = 1$$

Regrupieren.

$$x(1 - 8ce^x + 6a) + e^x(10c) - 12a + 2b = x + e^x$$

$$10c = 1 \rightarrow c = \frac{1}{10}$$

~~...~~

$$-8ce^x + 6a = 1$$

$$-\frac{8}{10}e^x + 6 = 1 \rightarrow -\frac{8}{10}e^x = -5$$

$$-12a + 2b = 0 \rightarrow -12 = -2b \rightarrow b = 6$$

$$y_p(x) = x^3 + 6x^2 + \frac{1}{10}x^2e^x$$

$$y_H = y_h + y_p = \underbrace{C_1 + C_2x + C_3e^x + C_4xe^x}_{y_h} + \underbrace{x^3 + 6x^2 + \frac{1}{10}x^2e^x}_{y_p}$$

$$10) \quad y^{(4)} + y^{(3)} = \cos 4x$$

Pas 1. Ec. caracter: $\pi^4 + \pi^3 = 0$

Horner

	π^4	π^3	π^2	π^1	π^0
	1	1	0	0	0

?

$$\pi^4 + \pi^3 = \pi^3(\pi + 1) = 0 \quad \rightarrow \quad \pi_1 = \pi_2 = \pi_3 = 0 - \text{rădăcină reală + triplet}$$

$$\pi_4 = -1 - \text{rădăcină reală distinctă}$$

$$y_1(x) = e^{\pi_1 x} = e^0 = 1$$

$$y_2(x) = x e^{\pi_2 x} = x e^0 = x$$

$$y_3(x) = x^2 e^{\pi_3 x} = x^2 e^0 = x^2$$

$$y_4(x) = e^{\pi_4 x} = e^{-x}$$

Se arată că $\forall (y_1, y_2, y_3, y_4) \neq 0 \rightarrow$

y_1, y_2, y_3, y_4 - linear independente,

y_1, y_2, y_3, y_4 - sistem fundamental de soluții.

$$y_0(x) = c_1 y_1(x) + c_2 y_2(x) + c_3 y_3(x) + c_4 y_4(x)$$

$$y_0(x) = c_1 + c_2 \cdot x + c_3 \cdot x^2 + c_4 \cdot e^{-x}, \quad c_1, c_2, c_3, c_4 - \text{constante.}$$

Pas 2. Căutăm soluție particulară.

$$f(x) = e^{\alpha x} [P(x) \cos(\beta x) + Q(x) \sin(\beta x)]$$

$$f(x) = e^0 [1 \cdot \cos(4 \cdot x) + 0 \cdot \sin(4 \cdot x)]$$

$$\rightarrow \alpha = 0 \quad P(x) = 1$$

$$\beta = 4 \quad Q(x) = 0$$

$\alpha + i\beta = 0 + 4i = 4i$ este rădăcină cu caracteristică ~~de~~ ordin 1.

$$y_p(x) = e^{\alpha x} [\cos(\beta x) P(x) + Q(x) \sin(\beta x)]$$

$$\mathcal{H}_{A,B} = \max\{g_1 P, g_2 Q\} = \max\{0, 0\} = 0$$

$$y_p(x) = A \cos(4x) + B \sin(4x)$$

$$= A \cos(4x) + B \sin(4x)$$

$$y_p'(x) = 4B \cos(4x) - 4A \sin(4x) = -4(A \sin(4x) - B \cos(4x))$$

$$y_p''(x) = -4(4B \sin(4x) + 4A \cos(4x)) = -16(B \sin(4x) + A \cos(4x))$$

$$y_p'''(x) = -16(4B \cos(4x) - 4A \sin(4x)) = 64(A \sin(4x) - B \cos(4x))$$

$$y_p^{(4)}(x) = 64(4B \sin(4x) + 4A \cos(4x)) = 256(B \sin(4x) + A \cos(4x))$$

$$256B \sin(4x) + 256A \cos(4x) + 64A \sin(4x) - 64B \cos(4x) = \cos 4x$$

$$= \cos 4x$$

$$\sin(4x)(256B + 64A) + \cos(4x)(256A - 64B) = \cos 4x$$

$$\begin{cases} 256B + 64A = 0 \\ 256A - 64B = 1 \end{cases} \quad \textcircled{+}$$

$$1088A = 1 \rightarrow A = \frac{1}{1088} = \frac{1}{272}$$

$$256B + 64 \cdot \frac{1}{272} = 0 \rightarrow B = -\frac{64}{272 \cdot \frac{272}{4}} = -\frac{1}{1088}$$

$$y_p(x) = \frac{1}{272} \cos 4x - \frac{1}{1088} \sin(4x)$$

$$y_G = y_0 + y_p = \underbrace{C_1 + C_2 x + C_3 x^2 + C_4 e^{-x}}_{y_0} + \underbrace{\frac{1}{272} \cos(4x) - \frac{1}{1088} \sin(4x)}_{y_p}$$

$C_1, C_2, C_3, C_4 \in \mathbb{R}$

Partea II.

/pag 1/

$$11) y'' - 2y' + y = \frac{e^x}{x^2 + 1}$$

Pas 1: $y'' - 2y' + y = 0$

Ec. caract: $\pi^2 - 2\pi + 1 = 0$

$$\Delta = b^2 - 4ac = 4 - 4 = 0$$

$$\pi_1 = \pi_2 = \frac{2}{2} = 1 \quad \text{- rădăcini reale duble}$$

$$y_1(x) = e^{\pi_1 x} = e^x$$

$$y_2(x) = x e^{\pi_2 x} = x e^x$$

$$y_2'(x) = e^x + x e^x$$

Se arată că $W(y_1, y_2) \neq 0 \rightarrow y_1, y_2$ - l.i., y_1, y_2 - sistem fundamental de soluții.

$$y_0(x) = C_1 \cdot e^x + C_2 \cdot x e^x, \quad C_1, C_2 \in \mathbb{R}$$

Pas 2:

$$C_1 \rightarrow C_1(x)$$

$$C_2 \rightarrow C_2(x)$$

Soluția generală a ec. neomogene este:

$$y(x) = C_1(x) \cdot e^x + C_2(x) \cdot x e^x$$

Determinăm pe $C_1(x), C_2(x)$ din sistemul:

$$\begin{cases} C_1' y_1 + C_2' y_2 = 0 \\ C_1' y_1 + C_2' y_2' = \frac{e^x}{x^2 + 1} \end{cases} \Rightarrow \begin{cases} C_1' e^x + C_2' x e^x = 0 \\ C_1' e^x + (e^x + x e^x) \cdot C_2' = \frac{e^x}{x^2 + 1} \end{cases} \quad | : e^x$$

$$\begin{cases} C_1' + x C_2' = 0 \rightarrow C_1' = -x C_2' \\ C_1' + (1+x) C_2' = \frac{1}{x^2 + 1} \end{cases}$$

$$\rightarrow (-x + 1 + x) C_2' = \frac{1}{x^2 + 1} \rightarrow C_2' = \frac{1}{x^2 + 1}$$

$$C_1'(x) = -x C_2'$$

$$C_2(x) = \int \frac{1}{x^2+1} dx = \arctan x + k_2.$$

$$C_1(x) = -\int \frac{2x}{x^2+1} dx = -\frac{1}{2} \int \frac{dt}{t} = -\frac{1}{2} \ln |t| = -\frac{1}{2} \ln |x^2+1| + k_1.$$

$$\begin{cases} t = x^2 + 1 \\ 2x dx = dt \end{cases}$$

$$\begin{aligned} y(x) &= \left(-\frac{1}{2} \ln |x^2+1| + k_1 \right) e^x + \left(\arctan x \cdot x^{k_2} \right) e^x, \quad k_1, k_2 \text{ constants} \\ &= k_1 \cdot e^x + k_2 x e^x - \frac{1}{2} \ln |x^2+1| \cdot \arctan x, \quad k_1, k_2 \text{ constants} \end{aligned}$$

Ecuații diferențiale de ordinul I.

$$1) \sqrt{1+x^2} y' + \sin y = 0$$

$$\sqrt{1+x^2} y' = -\sin y.$$

$$y' = \frac{-\sin y}{\sqrt{1+x^2}} = \sin y \cdot \frac{-1}{\sqrt{1+x^2}} \quad (1)$$

$$y' = \frac{dy}{dx}$$

$$\text{Din (1)} \rightarrow \frac{dy}{dx} = \sin y \cdot \frac{-1}{\sqrt{1+x^2}}$$

$$\frac{dy}{\sin y} = \frac{-1}{\sqrt{1+x^2}} dx \Rightarrow \int \frac{dy}{\sin y} = \int \frac{-1}{\sqrt{1+x^2}} dx$$

$$\int \frac{dy}{\sin y} = \int \frac{\frac{1}{2t}}{1+t^2} \cdot \frac{2}{1+t^2} dt = \int \frac{1}{t} dt = \ln|t| = \ln \left| \operatorname{tg} \left(\frac{y}{2} \right) \right|$$

Am aplicat substituția lui Weierstrass

$$t = \operatorname{tg} \left(\frac{y}{2} \right) \rightarrow \sin y = \frac{2t}{1+t^2}$$

$$dy = \frac{2}{1+t^2} dt$$

$$\begin{aligned} -\int \frac{1}{\sqrt{1+x^2}} dx &= -\ln \left(\left| x + \sqrt{x^2+1} \right| \right) + \ln c \\ &= \ln \left| \frac{1}{x + \sqrt{x^2+1}} + c \right|. \end{aligned}$$

Aplicăm arc tg. $\Rightarrow$

$$\Rightarrow \frac{y}{2} = \frac{1}{x + \sqrt{x^2+1}} + c \rightarrow y(x) = \frac{2}{x + \sqrt{x^2+1}} + 2c$$

$$2) 3y^2 y' = x(y^3 - 1)$$

$$y' = \frac{x(y^3 - 1)}{3y^2} \quad (1)$$

$$y' = \frac{dy}{dx}$$

$$\text{Ann (1)} \rightarrow \frac{dy}{dx} = \frac{x(y^3 - 1)}{3y^2}$$

$$\frac{\frac{dy}{(y^3 - 1)}}{\frac{3y^2}{3y^2}} = x dx \rightarrow \frac{3y^2}{y^3 - 1} dy = x dx \Rightarrow$$

$$\int \frac{3y^2}{y^3 - 1} dy = \int x dx$$

$$\begin{array}{l} y^3 - 1 = t \\ 3y^2 dy = dt \end{array} \rightarrow \int \frac{1}{t} dt = \frac{x^2}{2} + c.$$

$$\ln |t| = \frac{x^2}{2} + c, \text{ da } t = y^3 - 1.$$

$$\ln |y^3 - 1| = \frac{x^2}{2} + c$$

$$y^3 - 1 = e^{\frac{x^2}{2} + c}$$

$$y^3 = e^{\frac{x^2}{2} + c} - 1$$

$$y = \sqrt[3]{e^{\frac{x^2}{2} + c} - 1}$$

$$3) x^2(y+1) + (x^3-1)(y-1)y' = 0.$$

$$y' = \frac{-x^2(y+1)}{(x^3-1)(y-1)} \rightarrow y' = \frac{-x^2}{x^3-1} \cdot \frac{y+1}{y-1}.$$

$$\frac{dy}{dx} = \frac{-x^2}{x^3-1} \cdot \frac{y+1}{y-1}$$

$$\frac{y-1}{y+1} dy = -\frac{x^2}{x^3-1} dx.$$

$$\left(\frac{y+1}{y+1} - \frac{2}{y+1} \right) dy = -\frac{x^2}{x^3-1} dx \quad \int$$

$$\int \frac{y+1}{y+1} - 2 \int \frac{1}{y+1} dy = -\frac{1}{3} \int \frac{3x^2}{x^3-1} dx \quad \left\{ \begin{array}{l} t=x^3-1 \\ 3x^2 dx = dt \end{array} \right.$$

$$y - 2 \ln|y+1| = -\frac{1}{3} \ln|t| + \ln c.$$

$$y - 2 \ln|y+1| = -\frac{1}{3} \ln|t| + \ln c.$$

$$y - 2 \ln|y+1| = -\frac{1}{3} \ln|x^3-1| + \ln c.$$

$$y - 2 \ln|y+1| = -\frac{1}{3} \ln c(x^3-1)$$

$$4) \begin{cases} x^2 y' = x^2 + xy - y^2 \\ y(1) = 2 \end{cases}$$

$$x^2 y' = x^2 + xy - y^2$$

$$y' = \frac{x^2 + xy - y^2}{x^2} = \frac{x^2}{x^2} - \frac{xy}{x^2} - \frac{y^2}{x^2} = 1 + \frac{y}{x} - \frac{y^2}{x^2}$$

$$u = \frac{y}{x} \rightarrow y = u \cdot x \rightarrow y' = u' \cdot x + u$$

$$u' \cdot x + u = 1 + u - u^2$$

$$u' \cdot x = 1 - u^2 \rightarrow \frac{du}{1-u^2} = \frac{dx}{x} \xrightarrow{\int}$$

$$\int \frac{du}{1-u^2} = \int \frac{1}{x} dx \rightarrow \int \frac{du}{u^2-1} = \int \frac{dx}{x}$$

$$-\frac{1}{2} \ln \frac{u-1}{u+1} = \ln x + \ln c,$$

$$u = \frac{y}{x} \Rightarrow -\frac{1}{2} \ln \left(\frac{\frac{y}{x} - 1}{\frac{y}{x} + 1} \right) = \ln(c \cdot x) \quad | \cdot (-2)$$

$$\ln \left(\frac{\frac{y}{x} - 1}{\frac{y}{x} + 1} \right) = \ln(cx)^{-2}$$

$$y(1) = 2 \rightarrow \begin{matrix} x=1 \\ y=2 \end{matrix} \rightarrow \ln \left(\frac{\frac{2}{1} - 1}{\frac{2}{1} + 1} \right) = \ln c^{-2}$$

$$\frac{1}{3} = \frac{1}{c^2} \rightarrow c = \sqrt{3}$$

$$5) 2x^3 + (3x^2 + y^2)yy' = 0$$

$$2x^3 + (3x^2y + y^3)y' = 0$$

$$y' = \frac{-2x^3}{3x^2y + y^3}$$

$$\Rightarrow y' = \frac{-2x^3}{y^3(3\frac{x^2}{y^2} + 1)}$$

$$y' = \frac{-2\left(\frac{x}{y}\right)^3}{3\left(\frac{x}{y}\right)^2 + 1}$$

$$\frac{x}{y} = u \Rightarrow y = \frac{x}{u} \rightarrow y' = \frac{u - xu'}{u^2}$$

$$(u - xu') / (3u^2 + 1) = u^2 / (-2u^3)$$

$$3u^3 + u - 3xu^2 \cdot u' - xu' = -2u^5$$

$$(-3xu^2 - x) \cdot u' = -2u^5 - 3u^3 - u$$

$$u' = \frac{-2u^5 - 3u^3 - u}{-3xu^2 - x} \Rightarrow u' = \frac{-2u^5 - 3u^3 - u}{x(-3u^2 - 1)}$$

$$\frac{du}{dx} = \frac{1}{x} \cdot \frac{-2u^5 - 3u^3 - u}{-3u^2 - 1}$$

$$du \cdot \frac{-3u^2 - 1}{-2u^5 - 3u^3 - u} = \frac{1}{x} dx \quad \Rightarrow \int$$

$$\int \frac{-3u^2 - 1}{-2u^5 - 3u^3 - u} du = \ln x + \ln C$$

$$c) \begin{cases} y' = 4 + \frac{y}{x} + \left(\frac{y}{x}\right)^2 \\ y(1) = 2 \end{cases}$$

$$y = \frac{y}{x} \Rightarrow y = u \cdot x \Rightarrow y' = u' \cdot x + u$$

$$u' \cdot x + u = 4 + u + u^2$$

$$u' \cdot x = 4 + u^2$$

$$\frac{du}{dx} \cdot x = 4 + u^2 \Rightarrow \frac{du}{4 + u^2} = \frac{dx}{x} \Rightarrow$$

$$\int \frac{du}{u^2 + 4} = \int \frac{dx}{x} \Rightarrow \frac{1}{2} \arctg \frac{u}{2} = \ln x + C$$

$$u = \frac{y}{x} \rightarrow \frac{1}{2} \arctg \frac{y}{2x} = \ln x + C \quad | \cdot 2 \rightarrow$$

$$\arctg \frac{y}{2x} = \ln(Cx)^2 \xrightarrow[\text{tg}]{\text{Aplicam}}$$

$$\frac{y}{2x} = \text{tg}(\ln(C^2 x^2))$$

$$y(x) = 2x \text{tg}(\ln(C^2 x^2)) = 2x \text{tg}(\ln e^{\frac{y}{4}} x^2)$$

$$y(1) = 2 \rightarrow \begin{matrix} x=1 \\ y=2 \end{matrix} \rightarrow 2 = 2 \text{tg}(\ln C^2) \quad | : 2$$

$$1 = \text{tg} \ln C^2$$

$$\arctg 1 = \ln C^2$$

$$e^{\frac{1}{4}} = C^2 \rightarrow C = \sqrt{\frac{1}{e^{\frac{1}{4}}}}$$

$$7) y' = \frac{x+y+3}{2x+2y-1} = \frac{ax+by+c}{dx+ey+f}$$

$$a=1 \quad b=1 \quad c=3$$

$$d=2 \quad e=2 \quad f=-1$$

$$\frac{1}{2} = \frac{1}{2} \rightarrow dx \parallel dz$$

$$(x, y) \rightarrow (x, u), \quad u = u'(x)$$

$$u(x) = x+y \rightarrow u' = y'$$

$$u' = \frac{u+3}{2u-1} \quad - \text{ec. def. u var sep.}$$

$$u' = \frac{du}{dx} \rightarrow \frac{2u-1}{u+3} du = dx$$

$$\frac{2u}{u+3} - \frac{1}{u+3} = \frac{2(u+3)}{u+3} - \frac{6}{u+3} - \frac{1}{u+3}$$

$$2 \int \frac{u+3}{u+3} du - 6 \int \frac{1}{u+3} - \int \frac{1}{u+3} = \int x dx$$

$$2u - 6 \ln|u+3| - \ln|u+3| = x + C$$

$$u = x+y \Rightarrow$$

$$2x+2y - 6 \ln(x+y+3) - \ln(x+y+3) = x+C$$

$$2x+2y - \ln \frac{(x+y+3)^6}{x+y+3} = x+C$$

$$2x+2y - \ln(x+y+3)^5 = x+C$$

$$2x+2y - 5 \ln(x+y+3) = x+C$$

$$8) y' = \frac{x+y-2}{2x+y-3}$$

$$a=1 \quad b=1 \quad c=-2$$

$$d=2 \quad e=1 \quad h=-3$$

$$ae - bd = 1 - 2 = -1 \neq 0$$

$$x = X + A$$

$$y = Y + B$$

$$y' = \frac{dy}{dx} = \frac{dY}{dX}$$

$$\frac{dY}{dX} = \frac{x+A+Y+B-2}{2x+2A+Y+B-3} = \frac{X+Y}{2X+Y}$$

$$\begin{cases} A+B-2=0 & | \cdot (-2) \\ 2A+B-3=0 & \oplus \end{cases}$$

$$-B = -4 + 3 = -1 \rightarrow \underline{B=1}$$

$$2A - 2 = 0 \rightarrow 2A = 2 \rightarrow \underline{A=1}$$

$$\frac{dY}{dX} = \frac{x+y}{2x+y} = \frac{\cancel{x} \left(1 + \frac{y}{x}\right)}{x \left(2 + \frac{y}{x}\right)}$$

$$u = \frac{y}{x} \rightarrow y = u \cdot x \rightarrow y' = u' \cdot x + u$$

$$u' \cdot x + u = \frac{1+u}{2+u} \Rightarrow u'x = \frac{1+u}{2+u} - u \Rightarrow u'x = \frac{1+u-2-u}{2+u}$$

$$u'x = \frac{-1}{2+u} \quad | : x \rightarrow u' = \frac{-1}{2+u} \cdot \frac{1}{x}$$

$$u' = \frac{du}{dx} \Rightarrow \frac{du}{dx} = \frac{-1}{2+u} \cdot \frac{1}{x} \Rightarrow$$

$$\frac{1}{x} dx = \frac{2+u}{-1} du \Rightarrow$$

$$\int \frac{1}{x} dx = \int -2 - u du.$$

$$\ln|x| = -2u - \frac{u^2}{2} + C.$$

$$-4u - u^2 = 2 \ln x + C.$$

$$\text{Let } u = \frac{y}{x} \Downarrow$$

$$-4 \cdot \frac{y}{x} - \left(\frac{y}{x}\right)^2 = 2 \ln x + C.$$

$$g) x^2 y' - 2xy + 3 = 0 \quad | : x^2$$

$$y' - 2 \frac{y}{x} + \frac{3}{x^2} = 0 \rightarrow y' - \frac{2y}{x} = -\frac{3}{x^2}$$

$$P(x) = -\frac{2}{x}, \quad Q(x) = -\frac{3}{x^2}$$

$$y' = \frac{2}{x} y \Rightarrow \frac{dy}{dx} = \frac{2}{x} y \Rightarrow \frac{dy}{y} = \frac{2dx}{x} \xRightarrow{\int}$$

$$\Rightarrow \ln y = 2 \ln x + \ln C.$$

$$y_0 = e^{2 \ln x + C}$$

$$\text{Allgemein } y_N = k(x) \cdot e^{2 \ln x}$$

$$y_N' + P(x) y_N = Q(x).$$

$$k'(x) \cdot e^{2 \ln x} + k(x) \cdot \cancel{e^{2 \ln x}} \cdot 2 \cdot \frac{1}{x} - \frac{2}{x} \left(k(x) e^{2 \ln x} \right) = -\frac{3}{x^2}$$

$$k'(x) \cdot e^{2 \ln x} = \frac{-3}{x^2} \quad | : e^{2 \ln x}$$

$$k'(x) = \frac{-3}{x^2 \cdot e^{2 \ln x}} = \frac{-3}{x^2 \cdot x^2} = \frac{-3}{x^4}$$

$$k(x) = -3 \int \frac{1}{x^4} dx = -3 \left(-\frac{1}{3x^3} \right) = \frac{1}{x^3} + C.$$

$$y_N = \left(\frac{1}{x^3} + C \right) \cdot e^{2 \ln x}.$$

$$10) y' + 2xy = + 2xe^{-x^2}$$

$$P(x) = 2x$$

$$Q(x) = 2xe^{-x^2}$$

$$y' = -2xy$$

$$\frac{dy}{dx} = -2xy \rightarrow \frac{dy}{y} = -2x dx \xRightarrow{\int}$$

$$\int \frac{dy}{y} = -\int 2x dx \rightarrow \ln y = -\frac{2x^2}{2} + k$$

$$y = e^{-x^2 + k}$$

$$y_0 = e^{-x^2} + \frac{e^k}{e} = e^{-x^2} + e^{-1}$$

$$y_N = C(x) \cdot e^{-x^2}$$

$$y'_N + 2xy_N = 2xe^{-x^2}$$

$$(C(x) \cdot e^{-x^2})' + 2x C(x) \cdot e^{-x^2} = 2xe^{-x^2}$$

$$C'(x) \cdot e^{-x^2} + C(x) \cdot (-2x) \cdot e^{-x^2} + 2x C(x) \cdot e^{-x^2} = 2xe^{-x^2}$$

$$C'(x) \cdot e^{-x^2} = 2xe^{-x^2} \rightarrow C'(x) = 2x \rightarrow C(x) = \frac{2x^2}{2} = x^2 + k$$

$$y_N = (x^2 + k) \cdot e^{-x^2}$$

$$11) \begin{cases} y' \sin x - y = 1 - \cos x \\ y\left(\frac{\pi}{2}\right) = \frac{\pi}{2} \end{cases}$$

$$y' - \frac{1}{\sin x} y = \frac{1 - \cos x}{\sin x}$$

$$P(x) = -\frac{1}{\sin x}, \quad Q(x) = \frac{1 - \cos x}{\sin x}$$

$$y(x) = e^{-\int -\frac{1}{\sin x} dx} = e^{-\ln\left(\left|\tan \frac{x}{2}\right|\right)} = \left|\tan \frac{x}{2}\right|^{-1}$$

$$y(x) = \frac{1}{\left(\tan \frac{x}{2}\right)^{-1}} \left[\int \left(\tan \frac{x}{2}\right)^{-1} \cdot \tan \frac{x}{2} dx + C \right]$$

$$y(x) = \tan \frac{x}{2} (x + C)$$

$$y\left(\frac{\pi}{2}\right) = \frac{\pi}{2} \Rightarrow \frac{\pi}{2} = \tan \left(\frac{\pi}{4}\right) \left(\frac{\pi}{4} + C\right) \Rightarrow \frac{\pi}{2} = \frac{\pi}{4} + C$$

$$\frac{\pi}{2} = \frac{\pi}{4} + C \Rightarrow C = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$$

$$12) \quad xy' + y = xy^3$$

Ec. Bernoulli, $y' + P(x)y = Q(x)y^\alpha$

$$\alpha = 3$$

$$xy' + y = xy^3 \quad / : x$$

$$y' + \frac{y}{x} = y^3 \quad \Leftrightarrow \quad y' + P(x) \cdot y = Q(x) \cdot y^\alpha$$

Identificăm $P(x) = \frac{1}{x}$, $Q(x) = 1$, $\alpha = 3$.

$$I(x) = e^{\int (1-\alpha) P(x) dx} = e^{\int (1-3) P(x) dx} = e^{\int -2 \frac{1}{x} dx}$$

$$= e^{-2 \int \frac{1}{x} dx} = e^{-2 \ln x} = e^{\ln x^{-2}} = x^{-2} = \frac{1}{x^2}$$

$$y^{\alpha-1} = \frac{1}{I(x)} \left[\int (1-\alpha) Q(x) I(x) dx + c \right]$$

$$y^{-2} = \frac{1}{\frac{1}{x^2}} \left[\int -2 \cdot 1 \cdot \frac{1}{x^2} \cdot dx + c \right]$$

$$y^{-2} = x^2 \left(-2 \int \frac{1}{x^2} dx + c \right) = x^2 \left(-2 \cdot \frac{1}{x} + c \right)$$

$$\frac{1}{y^2} = \frac{-\frac{2x^2}{x} + x^2 c}{x^2}$$

$$y^2 = \frac{1}{-\frac{2x^2}{x} + x^2 c}$$

$$y(x) = \sqrt{\frac{1}{-2x + x^2 c}}$$

$$13) y' + \frac{y}{3} = e^x y^4$$

$$P(x) = \frac{1}{3}, \quad Q(x) = e^x, \quad \alpha = 4 = m$$

$$I(x) = e^{\int (1-m) \cdot P(x) dx} = e^{\int (1-4) \cdot \frac{1}{3} dx} = e^{\int -\frac{3}{3} dx} = e^{-x} = \frac{1}{e^x}$$

$$y^{(-3)} = \frac{1}{I(x)} \left[\int (1-m) Q(x) \cdot I(x) \cdot dx + c \right]$$

$$y^{-3} = \frac{1}{\frac{1}{e^x}} \left[\int -3 \cdot e^x \cdot \frac{1}{e^x} \cdot dx + c \right]$$

$$y^{-3} = e^x (-3x + c) = -3xe^x + e^x \cdot c$$

$$\frac{1}{y^3} = -3xe^x + e^x \cdot c$$

$$y^3 = \frac{1}{-3xe^x + e^x \cdot c} \rightarrow y(x) = \sqrt[3]{\frac{1}{-3xe^x + c \cdot e^x}}$$

$$14) xy' + 2y = -x^3 \cos x y^2 \quad | : y^2$$

$$x \frac{y'}{y^2} + 2 \cdot \frac{1}{y} = -x^3 \cos x$$

$$y = \frac{1}{y} \rightarrow u' = -\frac{1}{y^2} \cdot y'$$

$$-x \cdot u' + 2u = -x^3 \cos x \cdot \left(-\frac{1}{x}\right)$$

$$u' + \frac{2u}{x} = x^2 \cos x$$

$$u' - \frac{2u}{x} = 0$$

$$\frac{du}{dx} = \frac{2u}{x} \rightarrow \frac{du}{u} = 2 \frac{dx}{x} \Rightarrow \int$$

$$\int \frac{1}{u} du = 2 \int \frac{1}{x} dx \Rightarrow \ln u = 2 \ln x + \ln C$$

$$u_0 = x^2 C$$

$$u_N = x^2 C(x)$$

$$u_N = x^2 \cdot C(x)$$

$$(u_N)' + \left(-\frac{2}{x} u_N\right) = x^2 \cos x$$

$$2x \cdot C(x) + x^2 \cdot C'(x) - \frac{2}{x} \cdot x^2 \cdot C(x) = x^2 \cos x$$

$$x^2 C'(x) = x^2 \cos x$$

$$C'(x) = \cos x \rightarrow C(x) = \int \cos x dx = \sin x + C$$

$$u_N = x^2 (\sin x + C) = x^2 \sin x + x^2 C$$

$$\frac{1}{y} = x^2 \sin x + x^2 C \Rightarrow y = \frac{1}{x^2 \sin x + x^2 C}$$

$$15) \begin{cases} y' = y^2 - \frac{y}{x} - \frac{1}{x^2}, & y(1) = \frac{1}{x} \\ y(1) = 2 \end{cases}$$

$$\begin{cases} y' = y^2 - \frac{y}{x} - \frac{1}{x^2} \rightarrow y(1) = \frac{1}{x} \\ y(1) = 2 \end{cases}$$

$$y' = y^2 - \frac{y}{x} - \frac{1}{x^2} \quad \text{E.C. Riccati}$$

$$y' = P(x) \cdot y^2 + Q(x) \cdot y + R(x)$$

$$P(x) = 1, \quad Q(x) = -\frac{1}{x}, \quad R(x) = -\frac{1}{x^2}$$

$$D(x) \neq 0 \quad \text{DA} \rightarrow y(x) = S(x) + \frac{1}{x} = \frac{1}{x} + \frac{1}{x}$$

$$y'(x) = S'(x) + \left(\frac{1}{x}\right)' = -\frac{1}{x^2} - \frac{2}{x^2}$$

$$-\frac{1}{x^2} - \frac{2}{x^2} = \left(\frac{1}{x} + \frac{1}{x}\right)^2 - \frac{1}{x} \left(\frac{1}{x} + \frac{1}{x}\right) - \frac{1}{x^2}$$

$$-\frac{2}{x^2} = \cancel{\frac{1}{x^2}} + \frac{2}{x^2} + \frac{1}{x^2} - \cancel{\frac{1}{x^2}} - \frac{1}{x^2}$$

$$-\frac{2}{x^2} = \frac{1}{x^2} + \frac{1}{x^2} \neq 2 \rightarrow -2' = \frac{2}{x} + \frac{x}{1}$$

$$2' = -\frac{2+x}{x} \quad , \quad \dots$$

$$16) y' - y^2 = \frac{y}{x} + 9x^2 = 0, \quad y_p = 3x$$

$$y' = y^2 + \frac{1}{x} \cdot y - 9x^2 \quad \text{Ec. ricatti}$$

$$P(x) \neq 0.$$

$$y' = P(x) \cdot y^2 + \frac{1}{x} \cdot y - 9x^2$$

$$P(x) = 1, \quad Q(x) = \frac{1}{x}, \quad R(x) = -9x^2$$

$$S(x) = 3x \rightarrow y(x) = S(x) + \frac{1}{z} \rightarrow y' = S'(x) - \frac{z'}{z^2} = 3 - \frac{z'}{z^2}$$

$$3 - \frac{z'}{z^2} = \left(3x + \frac{1}{z}\right)^2 + \frac{1}{x} \left(3x + \frac{1}{z}\right) - 9x^2.$$

$$3 - \frac{z'}{z^2} = 9x^2 + \frac{6x}{z} + \frac{1}{z^2} + 3 + \frac{1}{zx} - 9x^2$$

$$\frac{z'}{z^2} = \frac{6x}{z} + \frac{1}{z^2} + \frac{1}{zx} \quad | \cdot z^2$$

$$-z' = 6zx + 1 + \frac{z}{x} \quad \text{Ec. liniară}$$

Rescriem ec. în variabila y .

$$-y' = 6xy + \frac{1}{x}y + 1 \quad | \cdot (-1)$$

$$y' = -6xy - \frac{1}{x}y - 1 \Rightarrow y' = y(-6x - \frac{1}{x}) - 1.$$

$$y' + y\left(6x + \frac{1}{x}\right) = -1.$$

$$14) y' - 3 + 3x^2y - xy^2, y_p = 3x$$

$$y' = -xy^2 + 3x^2y + 3 \quad \underline{\text{DAR}}$$

$$y' = P(x)y^2 + Q(x) \cdot y + R(x).$$

$$P(x) \neq 0 \rightarrow y(x) = S(x) + \frac{1}{z} = y = 3x + \frac{1}{z}.$$

$$y' = S'(x) + \left(\frac{-z'}{z^2} \right)$$

$$y' = 3 - \frac{z'}{z^2}.$$

$$3 - \frac{z'}{z^2} = -x \left(3x + \frac{1}{z} \right)^2 + 3x^2 \left(3x + \frac{1}{z} \right) + 3.$$

$$-\frac{z'}{z^2} = -x \left(9x^2 + \frac{6x}{z} + \frac{1}{z^2} \right) + 9x^3 + \frac{3x^2}{z}.$$

$$-\frac{z'}{z^2} = -9x^3 - \frac{6x^2}{z} - \frac{x}{z^2} + 9x^3 + \frac{3x^2}{z}.$$

$$-\frac{z'}{z^2} = -\frac{3x^2}{z} - \frac{x}{z^2} \cdot z^2$$

$$-z' = -3x^2z - x \quad | \cdot (-1)$$

$$z' = 3x^2z + x \quad \Leftrightarrow z' - 3x^2z = x$$

$$z' + P(x)z = Q(x).$$

$$z(x) = e^{-\int P(x)dx} \left(C + \int Q(x) \cdot e^{\int P(x)dx} dx \right) ?$$

↳ Rückwärt.

$$\text{Angenommen } y_N = k(x) \cdot e^{-\int P(x)dx}.$$

$$k \rightarrow k(x).$$

y_N - soll a. ec. normogene $\Rightarrow$

$$y_N' + P(x) \cdot y_N = Q(x).$$

$$\left[k(x) \cdot e^{-\int P(x) dx} \right]' + P(x) \cdot k(x) \cdot e^{-\int P(x) dx} = Q(x).$$

$$k'(x) \cdot e^{-\int P(x) dx} + \cancel{k(x) \cdot e^{-\int P(x) dx} \cdot [-P(x)]} + P(x) \cdot \cancel{k(x) \cdot e^{-\int P(x) dx}} = Q(x)$$

$$k'(x) \cdot e^{-\int P(x) dx} = Q(x)$$

$$k'(x) = e^{\int P(x) dx} \cdot Q(x).$$

$$k(x) = \int Q(x) \cdot e^{\int P(x) dx} dx + C.$$

$$y_N(x) = e^{-\int P(x) dx} \left[C + \int Q(x) \cdot e^{\int P(x) dx} dx \right].$$

$$y_N(x) = e^{\int 3x^2 dx} \left[C + \int x \cdot e^{\int 3x^2 dx} dx \right]$$

$$y_N(x) = e^{\frac{3x^3}{2}} \left[C + \int x \cdot e^{x^3} dx \right].$$